

Best Matches in the Context of Phylogenetic Networks

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1 Definitions and Notation

Sets and Maps Let X and S be finite sets. We consider maps $\sigma: X \rightarrow S$ assigning to each $x \in X$ a unique element $\sigma(x) \in S$ and where we always assume that σ is surjective. We will refer to the elements in S as *colors*. For a subset $Y \subseteq X$, we define $\sigma(Y) := \{\sigma(y) \mid y \in Y\}$ as the set of colors used for the elements in Y . For $x \in X$ and $s \in \sigma(X) \setminus \{\sigma(x)\}$, we define

$$X_s := \{y \in X \mid \sigma(y) = s\} \text{ and } X_{x,s} := \{x\} \cup X_s.$$

The power set of a given set X is denoted by 2^X . Two sets A and B *overlap* if $A \cap B \neq \emptyset$. A *clustering system* on X is a set $\mathcal{C} \subseteq 2^X$ such that (i) $\emptyset \notin \mathcal{C}$, (ii) $X \in \mathcal{C}$, and (iii) $\{x\} \in \mathcal{C}$ for all $x \in X$. The elements of a clustering system are called *clusters*. A clustering system is a *hierarchy*, if it does not contain pairwise overlapping clusters.

Let $(X, \ll)$ be totally ordered set and $\xi = (x_0, x_1, \dots, x_n)$ and $\xi' = (x'_0, x'_1, \dots, x'_n)$ be two distinct sequences of vertices in X . We say that ξ is *lexicographically smaller* than ξ' , in symbols, $\xi \ll_{\text{lex}} \xi'$, if there exists an index $k \in \{0, \dots, n\}$ such that $x_i = x'_i$ for all $i < k$, and $x_k \ll x'_k$. Moreover, for $a, b \in X$ we write $a \in \xi$ if there exists an index $i \in \{0, \dots, n\}$ such that $a = x_i$ and we write $[a, b] \in \xi$ if there exists an index $i \in \{0, \dots, n-1\}$ such that $a = x_i$ and $b = x_{i+1}$.

Digraphs and Colorings A *digraph* $G = (V, E)$ is an ordered pair consisting of a non-empty set $V(G) := V$ of *vertices* and a set $E(G) := E \subseteq (V \times V) \setminus \{(v, v) \mid v \in V\}$ of *arcs*. The set of out-neighbors of a vertex $x \in V$ in a digraph $G = (V, E)$ is given by $N_G^+(x) := \{v \in V \mid (x, v) \in E\}$. The *out-degree* of x in G is $|N_G^+(x)|$. A digraph $G = (V, E)$ is *bidirected* if $(x, y) \in E$ implies $(y, x) \in E$ for all $x, y \in V$.

An arc $e = (u, w)$ in a DAG G is a *shortcut* if there is a directed uw -path that does not contain the arc e [2, 5]. A DAG without shortcuts is *shortcut-free*. Throughout the paper, we will often use the “shortcut-free” version of a DAG G as defined next.

Definition 1.1. Let G be a DAG on X . Then, G^- denotes the DAG obtained from G by removal of all of its shortcuts.

Lemma 1.2. Let G be a DAG on X . Then, G^- is a DAG on X that is uniquely determined and shortcut-free. Moreover, $V(G) = V(G^-)$ and, for all $u, v \in V(G)$, we have $u \preceq_G v$ if and only if $u \preceq_{G^-} v$. In particular, if N is a network on X with root ρ so is N^- .

A colored digraph (G, σ) is a digraph equipped with a map $\sigma: V(G) \rightarrow S$. A colored digraph (G, σ) is *properly colored* if, for all $(u, v) \in E(G)$ it holds that $\sigma(u) \neq \sigma(v)$. For given colors $s_1, \dots, s_k \in \sigma(X)$, $k \geq 1$, we define $X_{s_1 \dots s_k} := \{x \in X \mid \sigma(x) = s_i \text{ for some } i \in \{1, \dots, k\}\}$. Moreover, $G_{s_1 \dots s_k}$ denotes as the k -colored induced subgraph of G and we define $\sigma_{s_1 \dots s_k} := X_{s_1 \dots s_k} \rightarrow \{s_1, \dots, s_k\}$ via $x \mapsto \sigma(x)$.

Definition 1.3 (sf-colored). Let $(G = (V, E), \sigma)$ be a colored digraph. The coloring σ is *sink-free* if it is proper, and for every vertex $v \in V$ and every color $s \in \sigma(V) \setminus \{\sigma(v)\}$, there is a vertex $u \in N_G^+(v)$ with $\sigma(u) = s$. A graph with a sink-free coloring is *sf-colored*.

Definition 1.4 (sicor-in-hub property). Let $(G = (V, E), \sigma)$ be a colored digraph. We call a vertex $v \in V$ a single color representative (sicor) if $\sigma(v) \neq \sigma(u)$ holds for all $u \in V \setminus \{v\}$. Moreover, a vertex $v \in V$ is an in-hub in (G, σ) if for all vertices $u \in V$ with $\sigma(u) \neq \sigma(v)$, it holds that $(u, v) \in E$. If each sicor vertex in (G, σ) is an in-hub, then (G, σ) has the sicor-in-hub property.

We remark that the existence of an sf-coloring, sicor, and the sicor-in-hub property can be checked for any vertex-colored graph (G, σ) in $O(|V| + |E|)$, i.e., linear, time.

Lemma 1.5. If a colored digraph (G, σ) is sf-colored, then (G, σ) is properly colored and has the sicor-in-hub property.

Proof. Let (G, σ) be an sf-colored digraph. By definition, (G, σ) is properly colored. If there exists no sicor $x \in V(G)$, then (G, σ) trivially satisfies the sicor-in-hub property. Assume that $x \in V(G)$ is a sicor. Hence, for all $y \in V \setminus \{x\}$, it holds that $\sigma(y) \neq \sigma(x)$. Since (G, σ) is sf-colored, for all $y \in V \setminus \{x\}$ there is a vertex $z \in N_G^+(y)$ with $\sigma(z) = \sigma(x)$. As x is a sicor, the latter two arguments imply that for all $y \in V \setminus \{x\}$, it holds that $\sigma(x) \neq \sigma(y)$ and $(y, x) \in E(G)$. Hence, x is an in-hub and (G, σ) has the sicor-in-hub property. $\square$

DAGs, Networks, LCAs and Leaf-Colorings. A directed $v_1 v_n$ -path $P = (V, E)$ has vertex set $V = \{v_1, v_2, \dots, v_n\}$ and the arcs in E are precisely of the form (v_i, v_{i+1}) , $i = 1, 2, \dots, n-1$. If there is an edge (v_n, v_1) and a directed $v_1 v_n$ -path in a digraph G , then G contains a cycle. A digraph without cycles is a directed acyclic graph (DAG).

We can associate a partial order $\preceq_G$ with a DAG G on the vertex set $V(G)$, defined by $v \preceq_G w$ if and only if there is a directed wv -path. In this case, we say that w is an ancestor of v and v is a descendant of w . If $v \preceq_G w$ and $v \neq w$, we write $v \prec_G w$. We sometimes use $w \succeq_G v$ instead of $v \preceq_G w$. If $(u, v) \in E(G)$, then u is a parent of v and v a child of u . Two vertices $u, v \in V(G)$ are $\preceq_G$ -incomparable, in symbols $u \parallel_G v$, if neither $u \preceq_G v$ nor $v \preceq_G u$ is true. We denote by $L(G) \subseteq V(G)$ the $\preceq_G$ -minimal vertices of G , and we call $x \in L(G)$ a leaf of G . Vertices $v \in V(G) \setminus L(G)$ are inner vertices of G . A vertex $v \in V(G)$ of a DAG G that is $\preceq_G$ -maximal is called a root, and the set of roots of G is denoted by $R(G)$. Note that $L(G) \neq \emptyset$ and $R(G) \neq \emptyset$ for all DAGs G [4]. A DAG G on X is a DAG with leaf-set X .

A (rooted) network N is a DAG for which $|R(N)| = 1$, i.e., N has a unique root $\rho \in V(N)$. We say that N is a network on X if $L(N) = X$. A (rooted) tree is a network that does not contain vertices v for which there are distinct arcs $(u, v), (u', v) \in E(N)$. A star-tree on X is a network on X with root ρ and arc set $\{(\rho, x) \mid x \in X\}$.

The leaf-set X of a network usually represents a set of genes, and we associate to each gene $x \in X$ the species $\sigma(x) \in S$ in which it resides. A network N on X equipped with such a map $\sigma: X \rightarrow S$ is denoted by (N, σ) and is called leaf-colored.

For a given network N and a subset $A \subseteq L(N)$, we denote with $\text{ANC}_N(A)$ the set of all vertices that are ancestors of all vertices contained in A in N . A $\preceq_N$ -minimal vertex v in $\text{ANC}_N(A)$ is a least common ancestor (LCA) of A in N . The set $\text{LCA}_N(A)$ comprises all LCAs of A in N . Since a network N has a unique root and since this root is a common ancestor for all $A \subseteq L(N)$, we obtain

Observation 1.6. If N is a network, then $\text{LCA}_N(A) \neq \emptyset$ for all $A \subseteq L(N)$.

If $|\text{LCA}_N(A)| = 1$, then $\text{LCA}_N(A) = \{v\}$ for some $v \in V(N)$ and we write $\text{lca}_N(A) = v$ and say the LCA of A is well-defined in this case. In a network N , however, it may happen that $|\text{LCA}_N(A)| > 1$. We are mainly concerned with LCAs for sets $A = \{x, y\}$. To simplify writing we use the notation $\text{lca}_N(x, y) := \text{lca}_N(\{x, y\})$ and $\text{LCA}_N(x, y) := \text{LCA}_N(\{x, y\})$.

Two DAGs G and H are inner-vertex disjoint if $(V(G) \setminus L(G)) \cap (V(H) \setminus L(H)) = \emptyset$, i.e., G and H may share some leaves but do not have any non-leaf vertices in common. Let $G_1, \dots, G_n$ be DAGs. Then, $G = \bigcup_{i=1}^n G_i$ is the digraph with vertex set $V(G) = \bigcup_{i=1}^n V(G_i)$ and arc set $E(G) = \bigcup_{i=1}^n E(G_i)$. In general, G is not a DAG. However, if the DAGs $G_1, \dots, G_n$ are pairwise inner-vertex disjoint, then G is a DAG in which LCAs are preserved as shown next.

Lemma 1.7. Suppose that $G_1, \dots, G_n$ are pairwise inner-vertex disjoint DAGs and put $G := \bigcup_{i=1}^n G_i$, $X := \bigcup_{i=1}^n L(G_i)$ and let I_{xy} be the set of all $i \in \{1, \dots, n\}$ for which $x, y \in L(G_i)$. Then, G is a DAG on X such that $\text{LCA}_G(x, y) = \bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y)$ for all $x, y \in X$.

Let $\rho_1, \dots, \rho_m$ be the roots of G , $m \geq 1$ and suppose that $G_1, \dots, G_n$ are networks. If $m = 1$, then put $N := G$ and, otherwise, add a vertex ρ to G and arcs (ρ, ρ_i) for all $i \in \{1, \dots, m\}$. Then, N is a network on X and $\bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y) = \text{LCA}_N(x, y)$ holds for all $x, y \in X$.

Proof. Let $G_1, \dots, G_n$ be pairwise inner-vertex disjoint DAGs and $G = \bigcup_{i=1}^n G_i$. It is straightforward to verify that G is a DAG on $X = \bigcup_{i=1}^n L(G_i)$. Now let $x, y \in X$ and I_{xy} be the set of all $i \in \{1, \dots, n\}$ for which $x, y \in L(G_i)$. Note that, if $x = y$, then $\text{LCA}_{G_i}(x, y) = \{x\}$ for all $i \in I_{xy}$. Since G is a DAG on X , it follows that $\text{LCA}_G(x, y) = \{x\}$. Hence, assume that $x \neq y$. We show first that $\text{LCA}_G(x, y) \subseteq \bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y)$. To this end, let $v \in \text{LCA}_G(x, y)$. Since $x \neq y$ it must hold that $v \notin X$. Moreover, by construction of G and the fact that $G_1, \dots, G_n$ are pairwise inner-vertex disjoint, it follows

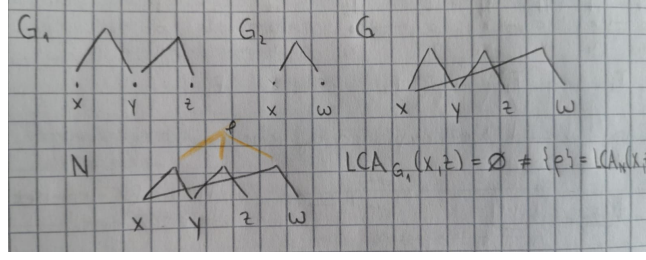


Figure 1: Network N for inner-vertex disjoint DAGs G_1 and G_2 with $\text{LCA}_{G_1}(x, y) \neq \text{LCA}_N(x, y)$.

that $v \in V(G_i)$ for exactly one $i \in \{1, \dots, n\}$. and that the vertex v is an ancestor of x and y in G_i . Suppose now, for contradiction, that $v \notin \text{LCA}_{G_i}(x, y)$. By definition, there exists $u \in \text{LCA}_{G_i}(x, y)$ such that $u \prec_{G_i} v$. Again, since $x \neq y$, we can conclude that $u \notin X$. Moreover, since $G_1, \dots, G_n$ are inner-vertex disjoint, u and v are located in the “copy” of G_i in G such that $u \prec_G v$ and u remains an ancestor of x and y ; a contradiction to $v \in \text{LCA}_G(x, y)$. Hence, $v \in \text{LCA}_{G_i}(x, y)$ and, thus, $v \in \bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y)$ must hold. We proceed with showing that $\bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y) \subseteq \text{LCA}_G(x, y)$. Assume that $v \in \bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y)$. Hence there is an $i \in I_{xy}$ such that $v \in \text{LCA}_{G_i}(x, y)$. Then, v is located in the “copy” of G_i in G and G_i is an induced subgraph of G that shares with any other “copies” G_j in G only leaves. It is now straightforward to verify that $v \in \text{LCA}_G(x, y)$. In summary, $\bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y) = \text{LCA}_G(x, y)$ holds.

Suppose now that $\rho_1, \dots, \rho_m$ are the roots of G , $m \geq 1$ and that $G_1, \dots, G_n$ are networks. Moreover, let N be as constructed in the statement of the lemma. One easily verifies that N is a network on X whose unique root is ρ (resp., ρ_1 in case $m = 1$). By Observation 1.6, $\text{LCA}_{G_i}(x, y) \neq \emptyset$ for all $x, y \in L(G_i)$ and all $i \in I_{xy}$. By the latter arguments, $\bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y) = \text{LCA}_G(x, y)$. The latter two arguments, in turn, imply that all leaves $x, y \in X$ have some LCA v that is located in G . This together with $v \prec_N \rho$ for all $v \in V(G)$ implies that $\rho \notin \text{LCA}_N(x, y)$ for all $x, y \in X$. In summary, $\text{LCA}_G(x, y) = \text{LCA}_N(x, y)$ and, therefore, $\bigcup_{i \in I_{xy}} \text{LCA}_{G_i}(x, y) = \text{LCA}_N(x, y)$. $\square$

We will often refer to such particular DAGs $G = \bigcup_{i=1}^n G_i$ as specified in Lemma 1.7 as *the DAG obtained from $G_1, \dots, G_n$ by identifying their leaves*. In Figure 1, an example is shown highlighting the importance of assuming that $G_1, \dots, G_n$ for the second statement.

Clusters, Regular Networks, and the Hasse diagram. For a network N on X and a vertex $v \in V(N)$, we put $C_N(v) = \{x \in X \mid x \preceq_N v\}$, i.e., the set of leaves that are descendants of v . This set is called the *cluster* of v , and we let $\mathfrak{C}_N = \{C_G(v) \mid v \in V(N)\}$ denote the set system on X comprising all clusters of N . Note that the set system $\mathfrak{C}_N$ is a clustering system for all networks N , cf. [3, L 14]. Moreover, following the definition of [3, Def 12], a network N has the path-cluster-comparability (PCC) property, if for all $u, v \in V(N)$ it holds that:

(PCC) u and v are $\preceq_N$ -comparable if and only if $C_N(u) \subseteq C_N(v)$ or $C_N(v) \subseteq C_N(u)$.

For later reference, we provide the following simple result.

Lemma 1.8. *A network N satisfies (PCC) if and only if its shortcut-free version N^- satisfies (PCC).*

Proof. Let N be a network on X and N^- is shortcut-free version. By Lemma 1.2, N and N^- are both networks on X and $u \preceq_N v$ if and only if $u \preceq_{N^-} v$ for all $u, v \in V(N)$. It is now straightforward to verify that $C_N(u) = C_{N^-}(u)$ must hold for all $u \in V(N) = V(N^-)$ and that N^- satisfies (PCC) precisely if N does. $\square$

By [3, L 24], for every clustering system $\mathfrak{C}$ there is a network N with $\mathfrak{C}_N = \mathfrak{C}$ that satisfies (PCC). This result builds on the concepts of Hasse diagrams and regular networks.

Given a set X , the *Hasse diagram* $\mathcal{H}(\mathfrak{C})$ of a set system $\mathfrak{C} \subseteq 2^X$ is the DAG with vertex set $\mathfrak{C}$ and an arc from $A \in \mathfrak{C}$ to $B \in \mathfrak{C}$ if (i) $B \subsetneq A$ and (ii) there is no $C \in \mathfrak{C}$ with $B \subsetneq C \subsetneq A$. We note that $\mathcal{H}(\mathfrak{C})$ is also known as the *cover digraph* of $\mathfrak{C}$ [1]. We now give a key definition.

Definition 1.9 ([1]). A DAG $G = (V, E)$ is regular if the map $\varphi: V \rightarrow V(\mathcal{H}[\mathfrak{C}_G])$ defined by $v \mapsto C_G(v)$ is an isomorphism between G and $\mathcal{H}(\mathfrak{C}_G)$.

Lemma 1.10 ([4, L 4.7]). Let $\mathfrak{C} \subseteq 2^X$ be a set system. Then, the Hasse diagram $\mathcal{H}(\mathfrak{C})$ is a shortcut-free DAG that satisfies (PCC). Moreover, if $\mathfrak{C}$ contains $\{x\}$ for all $x \in X$, then $\mathcal{H}(\mathfrak{C})$ is regular and phylogenetic. Furthermore, if $\mathfrak{C}$ is a clustering system, then $\mathcal{H}(\mathfrak{C})$ is a regular network.

Theorem 1.11 ([3, Thm 2]). *A network N is regular if and only if N does not contain vertices with out-degree 1, is shortcut-free, and satisfies (PCC).*

In general, we are interested in networks N with certain properties and that satisfy $\mathfrak{C}_N = \mathfrak{C}$ for a given clustering set system $\mathfrak{C}$. Structural properties of $\mathcal{H}(\mathfrak{C})$ are, in this context, often helpful. Although the Hasse diagram $\mathcal{H}(\mathfrak{C})$ of a clustering system $\mathfrak{C} \subseteq 2^X$ is always a network, it is neither a network on X nor does $\mathfrak{C}_{\mathcal{H}(\mathfrak{C})} = \mathfrak{C}$ hold, since the leaves of $\mathcal{H}(\mathfrak{C})$ are labeled with the inclusion-minimal elements in $\mathfrak{C}$, i.e., as sets. To circumvent this, we write

$$H \doteq \mathcal{H}(\mathfrak{C})$$

where H is the digraph obtained from $\mathcal{H}(\mathfrak{C})$ by relabeling all vertices $\{x\}$ in $\mathcal{H}(\mathfrak{C})$ by x . Thus, for $H \doteq \mathcal{H}(\mathfrak{C})$ it holds that $\mathfrak{C}_H = \mathfrak{C}$ provided that $\mathfrak{C}$ contains all singletons $\{x\}$ with $x \in X$

For later reference, we provide the following useful result.

Lemma 1.12. *Let N be a network on X and $H \doteq \mathcal{H}(\mathfrak{C}_N)$. If $v = \text{lca}_N(A)$ is well-defined in N for some $A \subseteq X$, then $\text{lca}_H(A)$ is well-defined in H and it holds that $v' = \text{lca}_N(A)$ for the vertex v' in H with $\mathfrak{C}_H(v') = \mathfrak{C}_N(v)$. In particular, if $\text{lca}_N(A)$ and $\text{lca}_N(B)$ are well-defined in N for some $A, B \subseteq X$ and $\text{lca}_N(A) \prec_N \text{lca}_N(B)$ (resp., $\text{lca}_N(A) = \text{lca}_N(B)$), then $\text{lca}_H(A) \prec_H \text{lca}_H(B)$ (resp., $\text{lca}_H(A) = \text{lca}_H(B)$).*

Proof. Let N be a network on X and $H \doteq \mathcal{H}(\mathfrak{C}_N)$. Suppose that $v = \text{lca}_N(A)$ is well-defined in N for some $A \subseteq X$. By construction of H , it holds that $\mathfrak{C}_H = \mathfrak{C}_N$. Thus, there is a unique vertex v' in H with $\mathfrak{C}_H(v') = \mathfrak{C}_N(v)$. Moreover, [3, Obs 12] implies that $\mathfrak{C}_N(v)$ is the unique inclusion-minimal cluster in $\mathfrak{C}_N$ containing A . As $\mathfrak{C}_H = \mathfrak{C}_N$ holds, $\mathfrak{C}_H(v')$ is the unique inclusion-minimal cluster in $\mathfrak{C}_H$ containing A . We now prove that $v' = \text{lca}_H(A)$. Assume that there is a cluster $\mathfrak{C} \in \mathfrak{C}_H \setminus \{\mathfrak{C}_H(v')\}$ such that $A \subseteq \mathfrak{C}$. Let w' be the vertex in H with $\mathfrak{C}_H(w') = \mathfrak{C}$. By the latter arguments, $\mathfrak{C}_H(v') \subsetneq \mathfrak{C}_H(w')$ follows. Moreover, by Lemma 1.10 and the fact that $\mathfrak{C}_N$ is a clustering system, Since H is regular. By Theorem 1.11, H satisfies (PCC). This together with $\mathfrak{C}_H(v') \subsetneq \mathfrak{C}_H(w')$ and [3, Obs 7] implies that $v' \prec_H w'$. Hence, for any vertex $w' \neq v'$ with $A \subseteq \mathfrak{C}_H(w')$ it holds that $v' \prec_H w'$. Consequently, $v' = \text{lca}_H(A)$.

Assume now that $\text{lca}_N(A)$ and $\text{lca}_N(B)$ are well-defined in N for some $A, B \subseteq X$. If $\text{lca}_N(A) = \text{lca}_N(B) = v$, then the latter arguments imply that $v' = \text{lca}_H(A) = \text{lca}_H(B)$ for the vertex v' in H with $\mathfrak{C}_H(v') = \mathfrak{C}_N(v)$.

Assume now that $v = \text{lca}_N(A) \prec_N \text{lca}_N(B) = u$. By the preceding arguments, there are vertices v' and u' in H with $\mathfrak{C}_H(v') = \mathfrak{C}_N(v)$ and $\mathfrak{C}_H(u') = \mathfrak{C}_N(u)$ that, in particular, satisfy $v' = \text{lca}_H(A)$ and $u' = \text{lca}_H(B)$. Since $v \prec_N u$, it holds that $\mathfrak{C}_N(v) \subseteq \mathfrak{C}_N(u)$ (cf. [3, L 17]). Moreover, by [4, L 3.1], $B \subseteq \mathfrak{C}_N(u)$ and there is no cluster $\mathfrak{C}_N(w)$ with $w \prec_N u$ such that $B \subseteq \mathfrak{C}_N(w)$. Hence, we obtain $B \not\subseteq \mathfrak{C}_N(v)$ and thus $\mathfrak{C}_N(v) \subsetneq \mathfrak{C}_N(u)$. It follows that $\mathfrak{C}_H(v') \subsetneq \mathfrak{C}_H(u')$. By Lemma 1.10, H is regular. Thus, we can apply Theorem 1.11 to conclude that H satisfies (PCC) which together with [3, Obs 7] implies that $v' \prec_H u'$. Hence, $\text{lca}_H(A) \prec_H \text{lca}_H(B)$. $\square$

2 Best Matches in Networks

Definition 2.1 (Best Matches). *Let (N, σ) be a leaf-colored network on X . A gene $y \in X$ is a best match of a gene $x \in X$ in (N, σ) , denoted $x \rightarrow y$, if $\sigma(x) \neq \sigma(y)$ and the following condition holds:*

For all $y' \in X$ with $\sigma(y') = \sigma(y)$ there is no $u \in \text{LCA}_N(x, y')$ such that $u \prec_N v$ for some $v \in \text{LCA}_N(x, y)$.

It is not difficult to verify that for a given leaf-colored network (N, σ) on X it holds that y is a best match of x for two leaves $x, y \in X$ with $\sigma(x) \neq \sigma(y)$ if and only if

$$u \not\prec_N v, \text{ or equivalently, } v \preceq_N u \text{ or } v \parallel_N u \quad (1)$$

for all $y' \in X$ with $\sigma(y') = \sigma(y)$ and all $u \in \text{LCA}(x, y')$ and all $v \in \text{LCA}(x, y)$.

Definition 2.2. *Let (N, σ) be a leaf-colored network on X . The digraph $\mathcal{G}(N, \sigma)$ with vertex set X , vertex coloring $\sigma: X \rightarrow S$, and with arcs (x, y) if and only if y is a best match of x in (N, σ) for all distinct $x, y \in X$ is the best match graph (BMG) of (N, σ) .*

An arbitrary colored digraph (G, σ) is a BMG if there exists some network (N, σ) with leaf set $L(N) = V(G)$ such that $(G, \sigma) = \mathcal{G}(N, \sigma)$. In these cases, we say that (G, σ) is explained by (N, σ) .

Remark 1. *To avoid trivial cases, we will always assume that $|\sigma(X)| > 1$ and thus, $|X| \geq 2$ throughout this paper.*

The aim of this paper is to understand, in more detail, which types of colored digraphs (G, σ) are BMGs.

As we shall see in Theorem 2.8, a colored digraph (G, σ) is a BMG if and only if (G, σ) is properly colored and has the sicor-in-hub property. To prove this result, we first define the following type of networks.

Definition 2.3. Two leaves x and y form a cherry $\{x, y\}$ in a network if they have the same parent. In a leaf-colored network (N, σ) , a cherry $\{x, y\}$ is a bicolored-cherry (BIC-cherry) if $\sigma(x) \neq \sigma(y)$.

A BIC-cherry network is a leaf-colored network (N, σ) on X with the following structure:

- If $|X| = 2$, then N has root $\rho = p_{xy}$ and the arcs (p_{xy}, x) , (p_{xy}, y) for $x, y \in X$, and no further vertices and arcs exist.
- If $|X| > 2$, then N has root ρ and, for any two leaves $x, y \in X$ with $\sigma(x) \neq \sigma(y)$, there exists a unique vertex p_{xy} and the arcs (ρ, p_{xy}) , (p_{xy}, x) , and (p_{xy}, y) , and there are no further vertices and arcs.

Note that, by Remark 1, $|X| \geq 2$ holds. In particular, in case $|X| = 2$, Remark 1 implies that $\sigma(x) \neq \sigma(y)$ for the vertices $x, y \in X$. In general, any two cherries x, y in a BIC-cherry network are, by definition, BIC-cherries.

Moreover, by Remark 1, $|\sigma(X)| > 1$, and thus, for every leaf $x \in X$ there is some vertex y with $\sigma(y) \neq \sigma(x)$ and thus, a vertex p_{xy} together with the arcs (ρ, p_{xy}) , (p_{xy}, x) . Hence, N is a DAG with unique root ρ and it follows that a BIC-cherry network as in Definition 2.3 is indeed a network on X . For an illustrative example of a BIC-cherry network, see Figure 2.

Lemma 2.4. Let (N, σ) be a BIC-cherry network on X . Then, for all $x, y \in X$ with $\sigma(x) \neq \sigma(y)$, it holds that $p_{xy} = \text{lca}_N(x, y)$ and that (x, y) and (y, x) is an arc in $\mathcal{G}(N, \sigma)$ and, if $|X| \geq 3$, then $\text{lca}_N(x, y) \neq \rho$.

Proof. Let (N, σ) be a BIC-cherry network on X . Observe first that, by construction a BIC-cherry network, the vertices p_{xy} and p_{ab} are always $\preceq_N$ -incomparable for all $x, y, a, b \in X$ with $\{x, y\} \neq \{a, b\}$. Moreover, since the only children of p_{xy} are the leaves x and y , it follows that $p_{xy} = \text{lca}_N(x, y)$. One easily verifies that, in case $|X| \geq 3$, we have $\text{lca}_N(x, y) \neq \rho$. By definition of best matches, (x, y) and (y, x) is an arc in $\mathcal{G}(N, \sigma)$ for $x, y \in X$ with $\sigma(x) \neq \sigma(y)$. $\square$

To obtain a network that can explain an arbitrary BMG (G, σ) , we will modify the BIC-cherry network and control the “non-arcs” in (G, σ) .

Definition 2.5. Let (N, σ) be a BIC-cherry network on X . The extension of N by $[xy : xz]$ for three leaves $x, y, z \in X$ is obtained by adding a new vertex q_{xz} and the arcs (p_{xy}, q_{xz}) , (q_{xz}, x) and (q_{xz}, z) . In this case, we say that x, y, z are part of an extension.

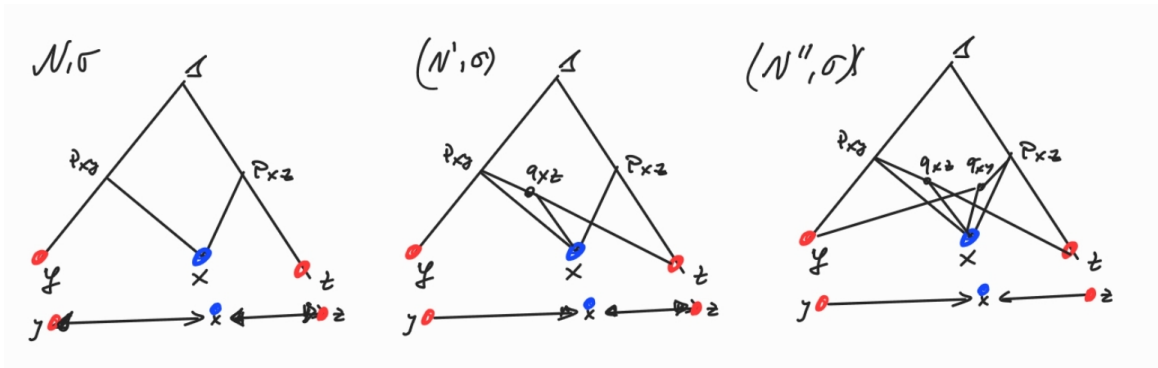


Figure 2: Left, a BIC-cherry network N with leaf set $X = \{x, y, z\}$ and leaf coloring $\sigma(x) \neq \sigma(y) = \sigma(z)$. The BMG $\mathcal{G}(N, \sigma)$ is shown directly below (N, σ) . In $\mathcal{G}(N, \sigma)$, all arcs between all pairs of vertices of different colors are present. Right, the BMG $\mathcal{G}(N', \sigma)$, drawn below (N', σ) , differs from $\mathcal{G}(N, \sigma)$ only by the absence of the single arc (x, y) . Here, N' coincides with the extension of N by $[xy : xz]$.

Lemma 2.6. Let (N, σ) be a BIC-cherry network on X with root ρ . Let N' be the digraph obtained from N by a finite number of extensions. Then the following properties hold.

- N' is a network on X with root ρ such that $\rho \notin \text{LCA}_{N'}(x, y)$ for all $x, y \in X$.
- If $x, y \in X$ are distinct leaves that are not part of an extension when constructing N' , then $\text{lca}_{N'}(x, y) = p_{xy}$.
- If an extension $[xy : xz]$ was applied, then $p_{xy} \in \text{LCA}_{N'}(x, y)$ and $q_{xz} \in \text{LCA}_{N'}(x, z)$.
- If, for some $x, y, z \in X$, there is a vertex $u \in \text{ANC}_{N'}(x, z)$ and a vertex $v \in \text{ANC}_{N'}(x, y) \setminus \{\rho\}$ with $u \prec_{N'} v$, then an extension $[xy : xz]$ was applied.

Proof. Let (N, σ) be a BIC-cherry network on X with root ρ . Since an extension requires the existence of at least three leaves $x, y, z \in X$, we assume in the following that $|X| \geq 3$. We prove that Properties (i), (ii), (iii), and (iv) are satisfied by induction on the number of extensions applied to obtain N' . As base case, no extension is applied, in which case $N' = N$ is clearly a network on X with root ρ . In particular, $x, y \prec_N p_{xy} \prec_N \rho$ holds for all $x, y \in X$. By the latter two arguments, Property (i) holds. By Lemma 2.4, Property (ii) holds. In addition, Property (iii) is vacuously true. Moreover, by construction of N , there is only one vertex v in N satisfying $x, y \prec_N v \prec_N \rho$ for all $x, y \in X$, namely $v = p_{xy}$. In other words, there are no vertices $u \in \text{ANC}_{N'}(x, z)$ and $v \in \text{ANC}_{N'}(x, y) \setminus \{\rho\}$ with $u \prec_N v$ for any $x, y, z \in X$, i.e., Property (iv) is vacuously true.

Suppose now that at least one extension step was applied to obtain N' . Let $\epsilon_1, \dots, \epsilon_n$ be the respective extension steps applied to N in this order, and suppose the properties are true for $k \geq 0$ extension steps applied to N . Let N_k be the digraph obtained from N after the first k extensions $\epsilon_1, \dots, \epsilon_k$. By assumption, N_k satisfies Properties (i), (ii), (iii), and (iv). Consider now the $k+1$ th extension $[xy : xz] := \epsilon_{k+1}$ and denote by N_{k+1} the digraph obtained from applying $[xy : xz]$ to N_k . Hence, we add a new vertex q_{xz} and the arcs (p_{xy}, q_{xz}) , (q_{xz}, x) and (q_{xz}, z) . Thus, only directed paths from p_{xy} to x, z are added and, in particular, no arcs of the form (w, ρ) or (l, w) with $w \in V(N_{k+1})$ and $l \in X$ have been added. It is now straightforward to verify that N_{k+1} remains a network on X with root ρ , i.e., N_{k+1} satisfies Property (i). Additionally, by the extension $[xy : xz]$, the vertices q_{xz} and p_{xy} become ancestors of x and z . The ancestor relationship for all other leaves $a \in X \setminus \{x, z\}$ remains unchanged in N_{k+1} compared to N_k . Hence, $\text{LCA}_{N_k}(a, b) = \text{LCA}_{N_{k+1}}(a, b)$ holds for all $a, b \in X$ with $\{a, b\} \neq \{x, z\}$. This, in particular, implies that new vertices q_{ab} introduced by previous extensions remain in $\text{LCA}_{N_{k+1}}(a, b)$. Thus, Property (ii) and $p_{xy} \in \text{LCA}_{N_{k+1}}(x, y)$ are satisfied by induction hypothesis. Since the only children of q_{xz} in N_{k+1} are x and z , it follows that $q_{xz} \in \text{LCA}_{N_{k+1}}(x, z)$. Hence, N_{k+1} satisfies Property (iii).

To verify Property (iv), let $x \in X$ be arbitrarily chosen. Suppose that, for some $y, z \in X$ there is a vertex $u \in \text{ANC}_{N_{k+1}}(x, z)$ and a vertex $v \in \text{ANC}_{N_{k+1}}(x, y) \setminus \{\rho\}$ with $u \prec_{N_{k+1}} v$. If this property already holds in N_k , then an extension $[xy : xz]$ was applied by induction hypothesis. Suppose that there are no vertices $u \in \text{ANC}_{N_k}(x, z)$ and $v \in \text{ANC}_{N_k}(x, y) \setminus \{\rho\}$ with $u \prec_{N_k} v$. It is easy to verify that, in this case, $\text{ANC}_{N_k}(x, y) = \{p_{xy}, \rho\}$ and that no extension $\epsilon_i = [xy : xz]$, $1 \leq i \leq k$ has been applied to obtain N_k . It now immediately follows that, in order to obtain N_{k+1} , the extension $\epsilon_{k+1} = [xy : xz]$ must have been applied. Thus, N_{k+1} satisfies Property (iv). In summary, N_{k+1} and consequently, $N' = N_n$ satisfies Property (i), (ii), (iii), and (iv). $\square$

Algorithm 1 Construction of network N explaining a BMG (G, σ)

Input: A properly colored digraph $(G = (X, E), \sigma)$ with sicor-in-hub property

Output: A leaf-colored network (N, σ) explaining (G, σ)

- 1: Let (N, σ) be the BIC-cherry network on X with root ρ .
 - 2: **for all** $x, y \in X$ with $\sigma(x) \neq \sigma(y)$ and $(x, y) \notin E$ **do**
 - 3: $N \leftarrow N$ extended by $[xy : xy']$ for some $y' \in X \setminus \{y\}$ with $\sigma(y) = \sigma(y')$
 - 4: **return** (N, σ)
-

Given a properly colored digraph (G, σ) with the sicor-in-hub property, we construct a leaf-colored network (N, σ) explaining (G, σ) by Algorithm 1. An example of the construction as in Algorithm 1 is illustrated in Figure 3. The correctness of Algorithm 1 is provided by the following:

Lemma 2.7. *For a properly colored digraph (G, σ) fulfilling the sicor-in-hub property, Algorithm 1 correctly returns a leaf-colored network (N, σ) explaining (G, σ) in $O(|V(G)|^2)$ time.*

Proof. Let $(G = (X, E), \sigma)$ be a properly colored digraph with the sicor-in-hub property. Assume first that $|X| = 2$. By Remark 1, $|\sigma(X)| = 2$ and thus $\sigma(x) \neq \sigma(y)$ for the two leaves $x, y \in X$. Hence, both x and y are sicors and, since (G, σ) has the sicor-in-hub property, both arcs (x, y) and (y, x) must exist in G . Consequently, the condition in the *for*-loop in Algorithm 1 is never satisfied and, thus, no extension is applied. It follows that (N, σ) returned by Algorithm 1 is precisely the BIC-cherry as specified in Definition 2.3 for the case $|X| = 2$. One easily verifies that (N, σ) explains (G, σ) .

Suppose now that $|X| \geq 3$. Observe first that whenever an extension $[xy : xy']$ in Algorithm 1 is applied, then x, y, y' are leaves of N . This together with Lemma 2.6 implies that the returned digraph N is a network on X with root ρ .

By definition of $\mathcal{G}(N, \sigma)$ and since N has leaf set X , the vertex set of $\mathcal{G}$ is X and, therefore, coincides with the vertex set of (G, σ) . Let $\tilde{E}$ be the arc set of $\mathcal{G}(N, \sigma)$. We prove now that $\mathcal{G}(N, \sigma) = (G, \sigma)$. To this end, it suffices to verify that $(x, y) \notin E$ if and only if $(x, y) \notin \tilde{E}$ for all $x, y \in X$.

Let $x, y \in X$ be two distinct arbitrarily chosen leaves. If $\sigma(x) = \sigma(y)$, then $(x, y) \notin E$, since G is properly colored and $(x, y) \notin \tilde{E}$ by Definition 2.2. Assume now that $\sigma(x) \neq \sigma(y)$.

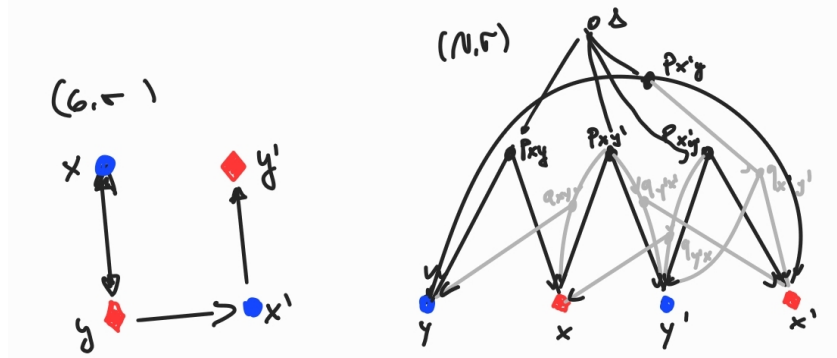


Figure 3: A BMG (G, σ) that is explained by (N, σ) . Extension of the underlying BIC-cherry network are highlighted in gray. Note that (G, σ) cannot be explained by a tree (T, σ) . To see this, assume that (G, σ) is explained by (T, σ) . In this case, $\text{lca}_T(y', x)$ and $\text{lca}_T(y', x')$ are well-defined and $\preceq_T$ -comparable, i.e., at least one of the vertices x and x' must be a best match of y' . However, no arc (y', x) and (y', x') exists in (G, σ) .

Suppose that $(x, y) \notin E$. Thus, y cannot be a sicor, since (G, σ) satisfies the sicor-in-hub property. Hence, there exists a $y' \in X$ with $\sigma(y) = \sigma(y')$ and the extension by $[xy : xy']$ as in Line 3 of Algorithm 1 is applied to the current network. Note that in the final network N returned by Algorithm 1, we have by Lemma 2.6, that $p_{xy} \in \text{LCA}_N(x, y)$ and $q_{xy'} \in \text{LCA}_N(x, y')$. Moreover, by construction, $q_{xy'} \prec_N p_{xy}$ holds. The latter arguments imply that y cannot be a best match of x . Hence, $(x, y) \notin \tilde{E}$.

Now assume that $(x, y) \notin \tilde{E}$. Hence, y is not a best match of x and it follows that there exists some $y' \in X$ with $\sigma(y) = \sigma(y')$ for which there is a vertex $u \in \text{LCA}_N(x, y')$ with $u \prec_N v$ for some $v \in \text{LCA}_N(x, y)$. Note that $p_{xy} \prec_N \rho$ and, hence, $\rho \notin \text{LCA}(x, y)$. Thus, there are vertices $u \in \text{ANC}_N(x, y')$ and $v \in \text{ANC}_N(x, y) \setminus \{\rho\}$ with $u \prec_N v$. By Lemma 2.6 (iv), an extension $[xy : xy']$ was applied in one of the steps by Line 3 of Algorithm 1. The execution of Line 3 requires, in particular, that $(x, y) \notin E$. In summary, $E = \tilde{E}$ and, therefore, $\mathcal{G}(N, \sigma) = (G, \sigma)$.

Finally, we show that Algorithm 1 can be implemented to run in $O(|X|^2)$. In the construction of the BIC-cherry network on X , we add for all $x, y \in X$ with $\sigma(x) \neq \sigma(y)$ a unique vertex p_{xy} and the arcs (ρ, p_{xy}) , (p_{xy}, x) , and (p_{xy}, y) . The latter task can be achieved in constant time for all fixed $x, y \in X$. As there are at most $O(|X|^2)$ distinctly colored vertices x, y in X , it follows that the initial BIC-cherry network on X in Line 1 can be constructed in $O(|X|^2)$ time. We now show that each individual check within the *for*-loop can be performed in constant time for every pair $x, y \in X$. Assuming that G is given as an adjacency matrix, verifying whether $(x, y) \notin E$ is clearly a constant-time operation. Likewise, checking whether $\sigma(x) \neq \sigma(y)$ can also be done in constant time. We now show that each individual check in Line 3 can be performed in constant time. To this end, assume that, in an initial pre-processing step preceding the *for*-loop, we scan all vertices in X and group them into color classes X_s , each containing all vertices $x \in X$ with $\sigma(x) = s$. For each vertex $x \in X$, we can determine $\sigma(x)$ in constant time and the set X_s to which the vertex x needs to be added in $O(|\sigma(X)|)$ time. Thus, establishing the color classes X_s for all $s \in \sigma(X)$ in a pre-processing requires $O(|\sigma(X)||X|) \subseteq O(|X|^2)$ time. Thereafter, selecting a vertex $y' \in X_{\sigma(y)} \setminus \{y\}$ is a constant-time operation. This is equivalent to selecting a vertex $y' \in X \setminus \{y\}$ with $\sigma(y') = \sigma(y)$, as required in Line 3. Moreover, the extension $[xy : xy']$ can clearly be carried out in constant time. Since the *for*-loop is executed at most $|X|^2$ times and each iteration, including the work in Line 3, can be completed in constant time, the total running time of the *for*-loop is $O(|X|^2)$. In summary, Algorithm 1 has time complexity $O(|X|^2)$. $\square$

Theorem 2.8. A colored digraph (G, σ) is a BMG if and only if (G, σ) is properly colored and has the sicor-in-hub property. In particular, it can be determined in $O(|V(G)|^2)$ time whether a given colored digraph (G, σ) is a BMG, and in the affirmative case, a network explaining (G, σ) can be constructed within the same time complexity.

Proof. If (G, σ) is a properly colored digraph with the sicor-in-hub property, then Lemma 2.7 implies that there is a leaf-colored network (N, σ) explaining (G, σ) . Hence, (G, σ) is a BMG.

Suppose now that (G, σ) is a BMG. Then there exists a leaf-colored network (N, σ) with root ρ explaining (G, σ) . Since N is a network, $\text{LCA}_N(x, y) \neq \emptyset$ for all $x, y \in X$. Suppose that $y \in X$ is a sicor in (G, σ) . In this case, there exists no $y' \in X \setminus \{y\}$ with $\sigma(y') = \sigma(y)$, and therefore, the leaf y is trivially a best match of x and $(x, y) \in E(G)$ for all $x \in X$ with $\sigma(x) \neq \sigma(y)$. Hence, (G, σ) has the sicor-in-hub property. Moreover, since arcs (x, y) exists in G only between best matches which requires $\sigma(x) \neq \sigma(y)$, it follows that G is properly colored.

As shown above, to test whether $(G = (X, E), \sigma)$ is a BMG, it suffices to verify that it is properly colored and satisfies the sicor-in-hub property. Since G contains $O(|X|^2)$ arcs, checking proper coloring by scanning all arcs (x, y) and verifying

$\sigma(x) \neq \sigma(y)$ requires $O(|X|^2)$ time. Next, we scan all vertices in X and group them into color classes $X_s = \{x \in X \mid \sigma(x) = s\}$ for all $s \in \sigma(X)$. As outlined in the proof of Lemma 2.7, this pre-processing step requires $O(|X|^2)$ time. For any vertex x , we can then determine in constant time whether it is a sicor by testing whether $|X_{\sigma(x)}| = 1$. If so, we check in $O(|X|)$ time whether $(y, x) \in E$ for all $y \in X \setminus \{x\}$. If this holds, x is a sicor-in-hub and the algorithm proceeds; otherwise, (G, σ) does not satisfy the sicor-in-hub property. Therefore, testing whether G has the sicor-in-hub property can be done in $O(|X|^2)$ time. Combined with the check for proper coloring, deciding whether (G, σ) is a BMG also takes $O(|X|^2)$ time. If (G, σ) is a BMG, and therefore properly colored and satisfying the sicor-in-hub property, Lemma 2.7 implies that a network explaining (G, σ) can be constructed in $O(|X|^2)$ time. This completes the proof. $\square$

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